Digital Twin Functional Framework

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Chapter 1 - Basic Functions and Details

1. Personal Profile and Identity Wallet

- **Description:** Central, encrypted profile containing verifiable attributes (name, age, driver's license, vaccination status, etc.).
- **User Control:** User owns the keys; can add, remove, and trigger selective sharing.
- **Security:** End-to-end encryption; local storage on the user's device as primary option; audit log for each sharing event.

2. Selective Disclosure

- **Description:** Share only the exact attributes a recipient needs (e.g., "over 18" without birthdate).
- **User Control:** Granular consent settings; time-limited sharing; option for one-time approval.
- **Security:** Principle of minimal disclosure; cryptographic proofs (verifiable credentials).

3. Consent and Policy Manager

- **Description:** Central overview of active consents, policies, and automated rules (e.g., who can see what in which situations).
- **User Control:** Easy withdrawal of consent; templates for quick approval; explanatory UI.
- **Security:** Timestamping, audit trails, option for third-party review.

4. Event-Triggered Verification (not passive monitoring)

- **Description:** Verification activated by specific events or user-initiated flows (e.g., boarding, payment, emergency).
- **User Control:** User chooses which events may trigger automatic verification.
- **Security:** No continuous biometric streaming without explicit consent; clear boundaries for data collection.

5. Emergency Signal and Controlled Emergency Disclosure (SOS)

- **Description:** Rapid activation of SOS flow that shares time-limited, relevant data with authorized responders (e.g., medical key data to ambulances).

- **User Control:** Predefined emergency profiles; manual or automatic activation (with prior user consent).
- **Security:** Strict access control; logging; option for user approval of shared data afterward.

6. Verifiable Certificates and Anti-Forgery

- **Description:** Digital certificates for documents, products, and qualifications, signed by issuer.
- **User Control:** User decides which certificates to present.
- **Security:** Public key infrastructures or decentralized verification; revocation mechanisms.

7. Routine and Preparation Assistant

- **Description:** Personal daily preparation function: meeting checklist, travel pre-check, medication reminders.
- **User Control:** User defines scope of automation; can toggle suggestions on/off.
- **Security:** Local data and anonymized models; no sharing without explicit consent.

8. Behavioral Insights and Anonymized Recommendations

- **Description:** Aggregated, anonymized insights for the user (e.g., sleep patterns, stress indicators) and concrete suggestions for improvement.
- **User Control:** User chooses which analyses may run; option to export or delete data.
- **Security:** Differential privacy or similar techniques when sharing aggregated data.

9. Integrations and APIs

- **Description:** Standardized, auditable APIs for healthcare systems, employers, authorities, and humanitarian organizations.
- **User Control:** Each integration requires explicit user approval; scopes and time limits.
- **Security:** OAuth-like flows, scopes, and regular security audits.

10. Recovery, Key Management, and Responsible Access

- **Description:** Procedures for account recovery, key backup, and emergency access (e.g., trusted contact).
- **User Control:** User defines trusted contacts and emergency scenarios.
- **Security:** Multi-party recovery, hardware-based keys as an option.

Chapter 2 - Additional Possibilities and Consent-Based Interventions

Introductory Principle: All interventions in this chapter require explicit, informed, and revocable consent. The system must never function as forced monitoring or replace professional responsibility.

A. Inmate and Rehabilitation Support

- **Purpose:** Support behavioral change, reduce isolation, and increase work readiness after release.
- **Examples:** Daily rhythm plans, structured learning modules, mentor contact, anonymized social network for support.
- **Consent/Governance:** Individual treatment plans prepared by professionals; consent for sharing with prison staff only in clearly defined, time-limited cases; independent complaint and review mechanism.
- **Safeguards:** Human oversight; clear limits on data sharing; psychological safety and exit options.

B. Sleep and Rhythm Coaching

- **Purpose:** Rapid adaptation to daily rhythms, improved sleep quality, and enhanced work function.
- **Examples:** Personal sleep plans, light and activity recommendations, behavioral nudges for sleep preparation.
- **Consent/Governance:** User chooses intervention level; clinically validated protocols recommended; data used only for improving user's own plan.
- **Safeguards:** No medical recommendations without professionals; referral to clinical treatment if needed.

C. Confidential Therapy Assistant

- **Purpose:** Prepare user for therapy, provide support between sessions, help maintain treatment goals.
- **Examples:** Conversation flows, reflection exercises, summaries for personal use, reminders of exercises.
- **Consent/Governance:** Conversations stored locally; sharing with therapist only after explicit agreement; clear boundaries on assistant's suggestions.
- **Safeguards:** Not a replacement for clinical treatment; signs of acute risk escalated to human professionals.

D. Workplace Adaptation and Onboarding

- **Purpose:** Faster adjustment to work rhythm, reduced absenteeism, increased retention.

- **Examples:** Personal workday plan, stress-prevention suggestions, social skills training modules, recommendations for shift work.
- **Consent/Governance:** Employee consent required; employer receives only aggregated, anonymized metrics with employee approval.
- **Safeguards:** Ban on forced monitoring; clear policies for data use in HR decisions.

E. Addiction Treatment and Behavioral Management

- **Purpose:** Support recovery, reduce relapse, and help manage adrenaline-driven patterns.
- **Examples:** Crisis plans, contact with support persons, alerts for risky behavior (only if user approves), structured exercises.
- **Consent/Governance:** Treatment plan prepared by professionals; user may withdraw consent at any time.
- **Safeguards:** Human clinical oversight; no automatic coercive mechanisms; emergency protocols with clear legal frameworks.

F. Behavioral Deviation and “Superego” Support in Critical Situations

- **Purpose:** When a treatment or behavioral plan is deviated from, provide support to bring user back on track.
- **Examples:** Gentle de-escalation, suggestions for next steps, contact with agreed support persons, temporary user-approved restrictions.
- **Consent/Governance:** Activated only with prior explicit consent; always with option for manual user override.
- **Safeguards:** Mandatory human review for any coercive intervention; legal and ethical compliance.

Implementation Principles and Recommended Guardrails

- **Consent First:** All functionality is based on informed, specific, and revocable consent. Default setting is “no sharing.”
- **Data Minimization:** Collect only what is necessary for the purpose; anonymize and aggregate where possible.
- **Human-in-the-Loop:** Clinical, legal, or safety-critical decisions require human approval.
- **Transparency and Audit:** User must be able to see who has access, when, and why; independent review recommended.
- **Ethical Review and Legal Compliance:** All intervention flows should be approved by an ethics board and comply with applicable law.

- **Measurement and Evaluation:** Define clear KPIs (e.g., reduced relapse, improved sleep score, faster work integration) and evaluate continuously.